Depth First Search

AlgoExpert · Easy · Graphs

The problem

- Given a tree-like graph of named nodes, return names in DFS (pre-order) order.

Intuition

- Depth-first search visits a node, then fully explores each child before moving to the next sibling.

Approach

- 1. Append the node's name
- 2. Recurse into each child in order
- 3. Return the collected list

Java solution

```java
public class DepthFirstSearch {
    public static class Node {
        public final String name;
        public final List<Node> children = new ArrayList<>();
        public Node(String name) { this.name = name; }
        public Node addChild(String name) { children.add(new Node(name)); return this; }
        public List<String> depthFirstSearch(List<String> out) {
            out.add(name);
            for (Node c : children) c.depthFirstSearch(out);
            return out;
        }
    }
}
```

Complexity

- Time: O(v + e)
- Space: O(v)

Recap

- Depth-first search visits a node, then fully explores each child before moving to the next sibling.
- Complexity: O(v + e)